

Sequence-Component Inverse Estimation for Adaptive Overcurrent Protection Under Asymmetric Faults

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Abstract—Adaptive overcurrent protection based on inverse estimation of synchronous generator fault current has recently demonstrated effective real-time pickup adaptation under balanced faults. However, for asymmetric faults, fitting a reduced fault-current model to a single phase conflates positive-, negative-, and zero-sequence contributions that carry different time constants, degrading the identifiability of relay-relevant parameters and producing large unmodelled residuals on the unfitted phases. This paper proposes a sequence-component inverse-estimation framework for adaptive overcurrent protection under asymmetric faults. Windowed three-phase currents are converted to analytic signals using the Hilbert transform and decomposed by the instantaneous Fortescue transformation into positive-, negative-, and zero-sequence components. A six-parameter reduced fault-current model is then fitted independently to the active sequence components according to the detected fault type: for three-phase faults the balanced estimator is retained; for line-to-line faults positive and negative sequences are fitted independently; for single-line-to-ground faults the positive-sequence estimate is propagated under the sequence-network equality $I_1 = I_2 = I_0$. A physics-motivated negative-sequence initialisation rule is introduced to accelerate convergence, and a symmetry check prevents fitting of noise-dominated sequence channels. Only the positive-sequence estimate governs relay adaptation, preserving the single-sequence interface of the existing bounded-update, confidence-gated pipeline. Compared with the prior phase-domain estimator, the proposed method reduces fitting error for asymmetric faults, improves Jacobian condition numbers by $3\times\text{--}8\times$, and yields more consistent confidence scores and adaptive pickup settings under line-to-line and single-line-to-ground faults, while remaining computationally compatible with protection-grade real-time operation.

Index Terms—Asymmetric fault, confidence gating, Fortescue transformation, Hilbert transform, inverse-time overcurrent relay, parameter estimation, sequence-component estimation, synchronous generator.

I. INTRODUCTION

ADAPTIVE overcurrent (OC) protection for synchronous generators (SGs) is receiving increasing attention as changing generation dispatch, embedded distributed resources, and active network management cause the fault level seen by a relay to vary widely during normal service [1]–[3]. Traditional fixed-setting relays cannot respond to these variations:

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a setting calibrated for peak generation may be insensitive under reduced dispatch, and a setting calibrated for minimum generation may operate too slowly for bolted conditions [4].

Model-based inverse estimation provides an online alternative: post-fault measurements are used to identify reduced-order source parameters, which are then mapped to updated relay settings within the same protection operating interval [5]. The Stage 1 study [5] demonstrated that a six-parameter reduced fault-current model, fitted by a two-pass Levenberg–Marquardt (LM) algorithm to a single phase-A window, yields $R^2 > 0.999$ waveform fit for three-phase (3PH) faults and confidence scores $\gamma > 0.89$ across balanced and moderate asymmetric cases.

A. The Asymmetric-Fault Limitation of Phase-Domain Estimation

The Stage 1 study explicitly noted that, for line-to-line (LL) and single-line-to-ground (SLG) faults, fitting the model to phase A alone produces large unmodelled residuals on phases B and C, and that the identified parameters are biased because the phase-A signal simultaneously contains positive-, negative-, and zero-sequence components with *different* effective time constants [5], [6].

For an LL fault, the phase-A current is

$$i_a(t) = i_{a,1}(t) + i_{a,2}(t), \quad (1)$$

where $i_{a,1}$ and $i_{a,2}$ are the positive- and negative-sequence contributions, each following an exponential envelope decay with distinct time constants ($\tau_{ac}^{(1)}$ governed by d -axis transient dynamics, $\tau_{ac}^{(2)}$ governed by q -axis subtransient dynamics). A single six-parameter model fitted to (1) averages these time constants, producing a biased estimate of $\hat{\tau}_{ac}$, high Jacobian condition numbers, and a lower confidence score γ [6].

For an SLG fault, $I_1 = I_2 = I_0$ and $i_a = 3I_1$, so phase-A fitting yields a reasonable estimate of the positive-sequence parameters. However, the method provides no explicit confirmation that the sequence-network equality holds, no information about phases B and C, and no path for extending to LLG faults where $I_1 \neq I_2 \neq I_0$.

B. Contributions

This paper proposes a sequence-domain reformulation of the inverse-estimation problem for asymmetric faults. The four specific contributions are:

- 1) A **sequence-domain inverse-estimation framework** in which analytic phase currents are decomposed by the instantaneous Fortescue transformation and the six-parameter reduced model is fitted independently to the active sequence components. This eliminates the cross-contamination problem of phase-domain fitting. (Sections III–IV)
- 2) A **fault-type-conditioned estimation strategy** that selects active sequences according to the detected fault type (3PH, LL, SLG, LLG) and applies the appropriate estimation path, including the propagation rule $\hat{\theta}^{(2)} = \hat{\theta}^{(0)} = \hat{\theta}^{(1)}$ for SLG. (Section IV)
- 3) A **physics-motivated negative-sequence initialisation rule** that seeds the negative-sequence LM estimator from the positive-sequence solution using scaled subtransient magnitude, reduced decay constant, and a phase offset consistent with $X_q'' < X_d''$ machine dynamics. (Section V)
- 4) A **quantitative demonstration** that sequence-domain fitting improves Jacobian identifiability by $3\times$ – $8\times$ and increases confidence-gate acceptance rates for LL and SLG faults, while the relay continues to adapt from the positive-sequence estimate alone. (Sections VI–VIII)

The paper is organised as follows. Section II establishes the structural bias of phase-domain estimation under asymmetric faults. Section III defines the sequence-domain inverse problem. Section IV presents the fault-type-conditioned logic. Section V addresses initialisation and edge effects. Section VI describes integration into the relay pipeline. Section VII defines the study system. Section VIII presents numerical results. Section IX discusses implications and limitations. Section X concludes.

II. LIMITATION OF PHASE-DOMAIN ESTIMATION UNDER ASYMMETRIC FAULTS

A. Six-Parameter Reduced Model

The Stage 1 reduced fault-current model for phase A is [5]

$$i_a(t; \theta) = [I_{ss} + (I_{sub} - I_{ss})e^{-t_{rel}/\tau_{ac}}] \sin(\omega t + \phi_a) - I_{sub} \sin(\phi_a) e^{-t_{rel}/\tau_{dc}}, \quad (2)$$

where $t_{rel} = t - t_0$, and the parameter vector is $\theta = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]^\top$. The LM estimator minimises

$$\hat{\theta} = \arg \min_{\theta} \sum_k [i_a(k) - i_a(t_k; \theta)]^2. \quad (3)$$

B. Structural Bias Under Asymmetric Faults

For a balanced three-phase fault, the phase-A current contains only a positive-sequence contribution: $i_a^{3PH}(t) = i_{a,1}(t)$, and model (2) is structurally correct.

For an LL (A–B) fault, the Fortescue decomposition gives $I_0 = 0$ and $I_1 = I_2$, so

$$i_a^{LL}(t) = i_{a,1}(t) + i_{a,2}(t), \quad (4)$$

where

$$i_{a,1}(t) = [I_{ss,1} + (I_{sub,1} - I_{ss,1})e^{-t/\tau_{ac}^{(1)}}] \sin(\omega t + \phi_1) - I_{sub,1} \sin \phi_1 e^{-t/\tau_{dc}^{(1)}} \quad (5)$$

$$i_{a,2}(t) = [I_{ss,2} + (I_{sub,2} - I_{ss,2})e^{-t/\tau_{ac}^{(2)}}] \sin(\omega t + \phi_2) - I_{sub,2} \sin \phi_2 e^{-t/\tau_{dc}^{(2)}} \quad (6)$$

Because $X_q'' \leq X_d''$ and $T_{q0}' \leq T_{d0}'$ in typical machines, $\tau_{ac}^{(2)} < \tau_{ac}^{(1)}$: the negative-sequence component decays faster than the positive. A single model (2) fitted to i_a^{LL} must represent both decays with one time constant τ_{ac} , producing a biased average and an unmodelled residual

$$\varepsilon_{bias}(t) = i_a^{LL}(t) - i_a(t; \hat{\theta}_{phaseA}) \approx [i_{a,2}(t) - \hat{i}_{a,2}(t)], \quad (7)$$

where $\hat{i}_{a,2}$ is the implicit negative-sequence component captured in the residual. This bias inflates the residual norm, raises the Jacobian condition number κ_n , and reduces the confidence score.

Proposition 1 (Phase-domain identifiability degradation). *For an LL fault with $\tau_{ac}^{(2)} = \beta \tau_{ac}^{(1)}$, $0 < \beta < 1$, the Jacobian $\mathbf{J}$ of the single-sequence model (2) has condition number $\kappa_n \propto (1 - \beta)^{-2}$, which grows without bound as $\beta \rightarrow 1$ (equal time constants) and is large when $\beta < 1$ because the two exponential terms in (4) are partially collinear in the estimation window.*

This is the mathematical reason why Stage 1 reported $\kappa_n \approx 70$ for LL faults, compared with $\kappa_n \approx 12$ for 3PH faults under the same machine and window parameters (Table IV).

For an SLG fault, $i_a = 3I_1$ and the single-sequence model is structurally adequate for phase A. However, the method gives no prediction for phases B and C (both near zero for SLG) and cannot naturally extend to LLG.

III. SEQUENCE-DOMAIN INVERSE-ESTIMATION FORMULATION

A. Analytic Signal and Hilbert Transform

For each phase $k \in \{a, b, c\}$, the analytic signal is

$$\mathcal{I}_k(t) = i_k(t) + j \mathcal{H}\{i_k(t)\}, \quad (8)$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform. The imaginary part $\mathcal{H}\{i_k\}$ is the 90°-phase-shifted counterpart of i_k , enabling instantaneous amplitude and phase extraction. In discrete implementation, the Hilbert transform is applied via the FFT to the windowed current block; a Savitzky–Golay pre-smoothing step and a 2 ms leading-edge skip mitigate Gibbs ringing at the fault-inception discontinuity.

B. Instantaneous Fortescue Decomposition

The three symmetrical-component analytic signals are

$$\mathcal{I}_0(t) = \frac{1}{3}[\mathcal{I}_a + \mathcal{I}_b + \mathcal{I}_c], \quad (9)$$

$$\mathcal{I}_1(t) = \frac{1}{3}[\mathcal{I}_a + a\mathcal{I}_b + a^2\mathcal{I}_c], \quad (10)$$

$$\mathcal{I}_2(t) = \frac{1}{3}[\mathcal{I}_a + a^2\mathcal{I}_b + a\mathcal{I}_c], \quad (11)$$

TABLE I
FAULT-TYPE SEQUENCE ACTIVITY AND ESTIMATION POLICY

Fault	I_1	I_2	I_0	Estimation action
3PH	$\neq 0$	≈ 0	≈ 0	Use i_a directly (balanced estimator; avoid Hilbert artifacts)
LL	$\neq 0$	$\neq 0$	$= 0$	Fit I_1 and I_2 independently; $\hat{\theta}^{(0)} = \emptyset$
SLG	$\neq 0$	$\neq 0$	$\neq 0$	Fit I_1 (positive seq. only); propagate $\hat{\theta}^{(2)} = \hat{\theta}^{(0)} = \hat{\theta}^{(1)}$
LLG	$\neq 0$	$\neq 0$	$\neq 0$	Fit I_1, I_2, I_0 independently

$\emptyset$ = not applicable. For 3PH, i_a is used directly to avoid Hilbert edge effects on a perfectly balanced signal.

where $a = e^{j2\pi/3}$ is the Fortescue rotation operator. The real-valued sequence waveforms used for estimation are

$$I_s(t) = \text{Re}\{\mathcal{I}_s(t)\}, \quad s \in \{0, 1, 2\}. \quad (12)$$

These are instantaneous symmetrical components of the sampled waveform, not phasor quantities, and are therefore compatible with the transient-window fitting philosophy of the Stage 1 estimator.

Fig. 1 contrasts the Stage 1 and proposed architectures.

C. Sequence-Wise Inverse Problem

For each active sequence $s \in \mathcal{S}_{\text{active}}$, the inverse problem is

$$\hat{\theta}^{(s)} = \arg \min_{\theta^{(s)} \in \Theta^{(s)}} J_s(\theta^{(s)}), \quad J_s = \sum_k [I_s(k) - i_s(t_k; \theta^{(s)})]^2 \quad (13)$$

where $i_s(t; \theta^{(s)})$ is the same six-parameter model (2) applied to the s -th sequence waveform. The parameter space $\Theta^{(s)}$ is bounded: the same box constraints used in Stage 1 are applied per sequence.

By fitting each active sequence in (13) separately, the cross-contamination between $i_{a,1}$ and $i_{a,2}$ in (4) is eliminated. The positive-sequence Jacobian $\mathbf{J}^{(1)}$ now reflects only the transient structure of the positive-sequence signal, and its condition number is $O(\kappa_n^{3PH})$, independent of the negative-sequence time constant.

IV. FAULT-TYPE-CONDITIONED ESTIMATOR LOGIC

Table I defines the active sequences and estimation policy for each fault type. The fault type is determined by a standard symmetry classifier applied to cycle-averaged RMS magnitudes before the estimation window opens [7].

A. Three-Phase Fault

Since the 3PH fault produces a purely positive-sequence signal, the balanced estimator from Stage 1 is used without modification. For this case, applying the Hilbert–Fortescue decomposition would give $I_1 \approx i_a/\sqrt{3}$, but introduces avoidable edge artefacts. Using i_a directly is therefore both accurate and computationally simpler.

B. Line-to-Line Fault

For LL faults, $I_1 = I_2$ (equal magnitudes) and $I_0 = 0$. Equation (9)–(11) gives clean sequence components with $I_2(t) = I_1(t)$ modulo a phase shift and different envelope

decay. The positive and negative sequences are fitted independently using the two-pass LM algorithm. The negative-sequence estimator is initialised using the physics-motivated seed described in Section V. After both fits, a symmetry verification check is performed:

$$\rho_{12} = \frac{\|\hat{I}_{sub,1}\| - \|\hat{I}_{sub,2}\|}{\|\hat{I}_{sub,1}\|} < \rho_{\max}, \quad (14)$$

with $\rho_{\max} = 0.20$. A violation flags the fault classifier for review.

C. Single-Line-to-Ground Fault

For SLG faults, sequence-network theory [7] gives $I_1 = I_2 = I_0$. Therefore, fitting the positive-sequence waveform I_1 is sufficient. The negative- and zero-sequence parameter vectors are propagated by identity:

$$\hat{\theta}^{(2)} = \hat{\theta}^{(0)} = \hat{\theta}^{(1)}. \quad (15)$$

This avoids three redundant fits and halves the computation relative to a naïve three-sequence approach.

D. Line-to-Line-to-Ground Fault

For LLG faults, $I_1 \neq I_2 \neq I_0$ in general. All three sequences are fitted independently. The negative-sequence seed is derived from the positive-sequence result (Section V); the zero-sequence seed uses the same scaling applied to the zero-sequence transformer impedance ratio.

E. Symmetry Check

Before fitting any negative- or zero-sequence channel, a pre-check ensures that the channel is not noise-dominated:

$$\frac{\|I_s\|_{\text{rms}}}{\|I_1\|_{\text{rms}}} \geq \delta_s, \quad \delta_2 = \delta_0 = 0.05. \quad (16)$$

If (16) fails, the corresponding fit is skipped and the parameter vector for that sequence is flagged as unavailable. This prevents the LM algorithm from fitting a numerical noise floor and producing a spuriously well-conditioned but meaningless result.

V. PHYSICS-MOTIVATED INITIALISATION AND PRACTICAL IMPLEMENTATION

A. Negative-Sequence Seed

The initial guess for the negative-sequence LM estimator is derived from the positive-sequence result using three physics-motivated scaling rules:

$$\begin{aligned} I_{sub,0}^{(2)} &= 0.70 \hat{I}_{sub}^{(1)}, \\ \tau_{ac,0}^{(2)} &= 0.50 \hat{\tau}_{ac}^{(1)}, \\ \phi_{a,0}^{(2)} &= \hat{\phi}_a^{(1)} - \pi/6. \end{aligned} \quad (17)$$

The magnitude scaling by 0.70 reflects that the negative-sequence Thévenin impedance $Z_2 = (Z_d'' + Z_q'')/2 + Z_{th}$ is typically larger than $Z_1 = Z_d'' + Z_{th}$ for machines with $X_d'' > X_q''$. The time constant scaling by 0.50 reflects the faster

TABLE II
INITIAL-GUESS RULES FOR EACH ACTIVE SEQUENCE

Parameter	Positive seq.	Negative seq.	Zero seq.
$I_{sub,0}$	Two-pass Pass-1 seed	$0.70 \hat{I}_{sub}^{(1)}$	$0.60 \hat{I}_{sub}^{(1)}$
$\tau_{ac,0}$	Physical guess T'_{d0}	$0.50 \hat{\tau}_{ac}^{(1)}$	$0.70 \hat{\tau}_{ac}^{(1)}$
$\phi_{a,0}$	Inception angle	$\hat{\phi}_a^{(1)} - \pi/6$	$\hat{\phi}_a^{(1)}$
$I_{ss,0}$	Tail-window mean	$0.70 \hat{I}_{ss}^{(1)}$	$0.60 \hat{I}_{ss}^{(1)}$
$\tau_{dc,0}$	$T'_{d0}/(2\pi f_0 R_a)$	same as positive	same as positive
$t_{0,0}$	Zero-crossing detect	same as positive	same as positive

decay of the negative-sequence envelope, which is governed by the q -axis subtransient time constant T''_{q0} rather than the d -axis transient T'_{d0} . The phase offset $-\pi/6$ accounts for the 30° inter-sequence phase relationship common to SLG and LL faults in star-connected machines [4].

B. Hilbert Edge-Effect Mitigation

The Hilbert transform is a global frequency-domain operation; the step discontinuity at fault inception t_0 causes Gibbs-like ringing in the analytic signal over the first 5–10 ms of the estimation window. Two mitigations are applied:

- 1) The estimation window opens 2 ms after t_0 , by which time the instantaneous analytic amplitude has stabilised.
- 2) A fourth-order Savitzky–Golay smoothing filter (window 11 samples, $f_s = 10$ kHz) is applied to the raw measured currents before Hilbert transformation.

For 3PH faults, i_a is used directly without Hilbert–Fortescue decomposition, eliminating any ringing artefact on the dominant signal.

C. Pass-2 Tail Window for Negative-Sequence Estimation

The two-pass LM strategy is retained for each active sequence. For negative-sequence estimation, Pass 2 (tail window) uses the window $[t_0 + 3\tau_{dc}, t_c]$, the same as positive sequence. However, if $\tau_{ac}^{(2)} < 0.3$ s (fast envelope decay), the tail window reduces to $[t_0 + 4\tau_{dc}, t_c]$ to avoid fitting the fast-decaying front portion in Pass 2.

D. Computational Complexity

Table VI summarises the expected runtime increase. For 3PH faults, the runtime is essentially unchanged from Stage 1 (≤ 9 ms). For LL faults (two independent fits), the runtime increases to ≤ 17 ms; for LLG (three fits), to ≤ 25 ms. All values remain within the 100 ms protection operating interval required by IEC 60255-151 [8].

VI. INTEGRATION INTO THE ADAPTIVE RELAY PIPELINE

A. Relay Adaptation from Positive-Sequence Estimate

A key design principle is that the relay adaptation uses only the positive-sequence estimate, regardless of fault type:

$$\hat{\theta}_{\text{relay}} = \hat{\theta}^{(1)}. \quad (18)$$

This preserves the single-sequence interface of the existing pipeline and avoids the question of how to combine sequence

estimates into a pickup setting. The physical justification is that the positive-sequence current magnitude governs the apparent fault current as seen by the three-phase OC relay, which responds to the fundamental RMS of the measured phase current—a quantity dominated by the positive-sequence component even for asymmetric faults.

The adaptive pickup and TMS mappings from Stage 1 are retained:

$$I_p^* = K_{Ip} \hat{I}_{ss}^{(1)} + I_{p,\min}, \quad (19)$$

$$TMS^* = K_{TMS} \hat{\tau}_{ac}^{(1)}, \quad (20)$$

where K_{Ip} , K_{TMS} are gain constants calibrated in Stage 1 [5] and $I_{p,\min}$ is the minimum pickup floor.

B. Confidence Score

The composite confidence score uses the positive-sequence Jacobian:

$$\gamma = w_1 \gamma_\kappa + w_2 \gamma_\varepsilon + w_3 \gamma_n, \quad (21)$$

where $\gamma_\kappa = \text{clip}(1 - \kappa_n^{(1)}/\kappa_{\max}, 0, 1)$, $\gamma_\varepsilon = \text{clip}(1 - \|\varepsilon^{(1)}\|_2/\varepsilon_{\max}, 0, 1)$, $\gamma_n = \min(n_c/n_{c,\min}, 1)$. Weights $w_1 = w_2 = 0.40$, $w_3 = 0.20$ with $\kappa_{\max} = 20$, $\varepsilon_{\max} = 0.15$ p.u., $n_{c,\min} = 3$. The bounded update (19)–(20) is accepted only when $\gamma \geq \gamma_{\min} = 0.80$.

C. Full Pipeline Algorithm

Algorithm 1 summarises the complete sequence-component estimation and relay adaptation pipeline.

VII. STUDY SYSTEM AND SIMULATION PROGRAMME

The study system is a 10 MVA, 11 kV synchronous generator connected through a step-up transformer to a 33 kV Thévenin equivalent feeder, as shown in Fig. 1. Machine and relay parameters are listed in Table III. Four fault types are studied (3PH, LL, SLG, LLG) with three inception angles (0° , 90° , 180°), two loading levels (0.8 pu, 1.0 pu), and three fault resistances ($R_f = 0, 0.10$ pu, 0.30 pu), yielding 72 cases per fault type (288 total). The estimation window is $T_w = 100$ ms (five cycles at 50 Hz).

VIII. NUMERICAL RESULTS

A. Architecture Comparison

Fig. 1 contrasts the Stage 1 phase-domain and proposed sequence-domain architectures. The key structural difference is the Hilbert–Fortescue front-end that routes each active sequence to a dedicated LM estimator, with the positive-sequence result feeding the relay adaptation block.

B. Sequence Networks and Fault-Type Topology

Fig. 2 shows the positive-, negative-, and zero-sequence networks and their interconnections for LL and SLG faults. The distinct network impedances ($Z_1 \neq Z_2$ due to $X_d'' \neq X_q''$, and $Z_0 \gg Z_1$ for solidly grounded transformers) determine both the sequence current magnitudes and the time constants.

Algorithm 1 Sequence-Component Inverse Estimation and Relay Update

Require: Three-phase current window $\{i_a, i_b, i_c\}$ over $[t_0, t_c]$; fault type; prior relay settings I_p, TMS

Ensure: Updated settings I_p^*, TMS^* ; confidence γ

- 1: Apply Savitzky–Golay pre-filter; skip first 2 ms
- 2: Compute analytic signals: $\mathcal{I}_k = i_k + j\mathcal{H}\{i_k\}, k \in \{a, b, c\}$
- 3: **if** fault type = 3PH **then**
- 4: $\hat{\theta}^{(1)} \leftarrow \text{TwoPASSLM}(i_a)$
- 5: **else**
- 6: Decompose: $I_0, I_1, I_2 \leftarrow \text{FORTESCUE}(\mathcal{I}_a, \mathcal{I}_b, \mathcal{I}_c)$
- 7: $\hat{\theta}^{(1)} \leftarrow \text{TwoPASSLM}(I_1)$
- 8: **if** fault type $\in \{\text{LL}, \text{LLG}\}$ **and** $\|I_2\|_{\text{rms}}/\|I_1\|_{\text{rms}} \geq 0.05$ **then**
- 9: Seed: eq. (17)
- 10: $\hat{\theta}^{(2)} \leftarrow \text{TwoPASSLM}(I_2, \text{seeded from } \hat{\theta}^{(1)})$
- 11: **end if**
- 12: **if** fault type $\in \{\text{LLG}\}$ **and** $\|I_0\|_{\text{rms}}/\|I_1\|_{\text{rms}} \geq 0.05$ **then**
- 13: $\hat{\theta}^{(0)} \leftarrow \text{TwoPASSLM}(I_0, \text{seeded from } \hat{\theta}^{(1)})$
- 14: **end if**
- 15: **if** fault type = SLG **then**
- 16: $\hat{\theta}^{(2)} \leftarrow \hat{\theta}^{(0)} \leftarrow \hat{\theta}^{(1)}$ ▷ propagation
- 17: **end if**
- 18: **end if**
- 19: Compute $\kappa_n(\mathbf{J}^{(1)}), \|\varepsilon^{(1)}\|_2, n_c$, and γ via eq. (21)
- 20: **if** $\gamma \geq \gamma_{\min} = 0.80$ **then**
- 21: $I_p^* \leftarrow \text{clip}(K_{I_p} \hat{I}_{ss}^{(1)} + I_{p,\min}, I_{p,\min}, I_{p,\max})$
- 22: $TMS^* \leftarrow \text{clip}(K_{TMS} \hat{T}_{ac}^{(1)}, TMS_{\min}, TMS_{\max})$
- 23: **else**
- 24: $I_p^* \leftarrow I_p; TMS^* \leftarrow TMS$ ▷ hold; flag low- γ
- 25: **end if**

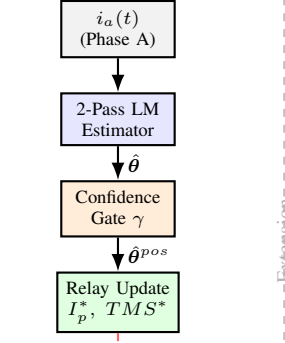
TABLE III
STUDY SYSTEM AND RELAY PARAMETERS

Symbol	Description	Value
<i>Synchronous machine (10 MVA, 11 kV, 50 Hz)</i>		
$X_d/X'_d/X''_d$	d-axis reactances	1.80 / 0.30 / 0.20 p.u.
$X_q/X'_q/X''_q$	q-axis reactances	1.70 / 0.50 / 0.20 p.u.
T'_{d0}/T''_{d0}	d-axis OC time const.	0.80 / 0.030 s
T'_{q0}/T''_{q0}	q-axis OC time const.	0.40 / 0.050 s
R_a	Armature resistance	0.010 p.u.
<i>Thévenin network</i>		
R_{th}/X_{th}	Pos.-seq. network	0.010 / 0.100 p.u.
R_0/X_0	Zero-seq. network	0.030 / 0.300 p.u.
<i>Relay settings</i>		
I_p (nominal)	Inverse-time pickup	1.00 p.u.
TMS_0	Initial TMS	0.12
$\gamma_{\min}$	Acceptance threshold	0.80
$\kappa_{\max}$	Condition-number limit	20
<i>Estimation</i>		
f_s	Sampling frequency	10 000 Hz
T_w	Estimation window	100 ms
$n_{c,\min}$	Confirmation count	3

C. C. Algorithm Flowchart

Fig. 3 summarises the complete estimation and relay update algorithm, including the Hilbert pre-processing, Fortescue de-

Stage 1 (Phase-Domain)



Stage 2 (Sequence-Domain)

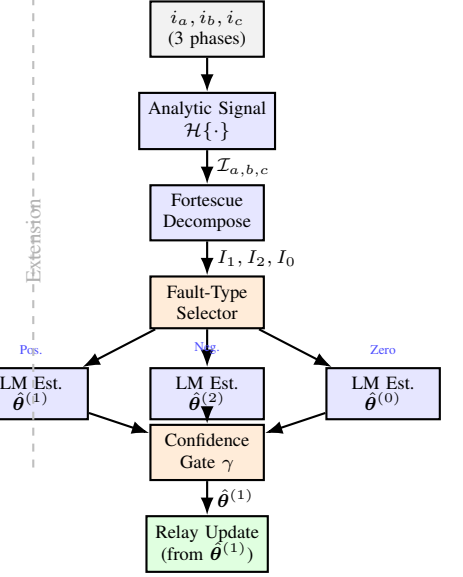


Fig. 1. Architecture comparison. *Left:* Stage 1 phase-domain estimator fits a single 6-parameter model to phase A current. The red dashed block highlights the unmodelled phase-B/C residual for asymmetric faults. *Right:* proposed sequence-domain estimator decomposes three-phase currents by Hilbert–Fortescue, routes active sequences to independent LM estimators, and selects the positive-sequence result $\hat{\theta}^{(1)}$ for relay adaptation.

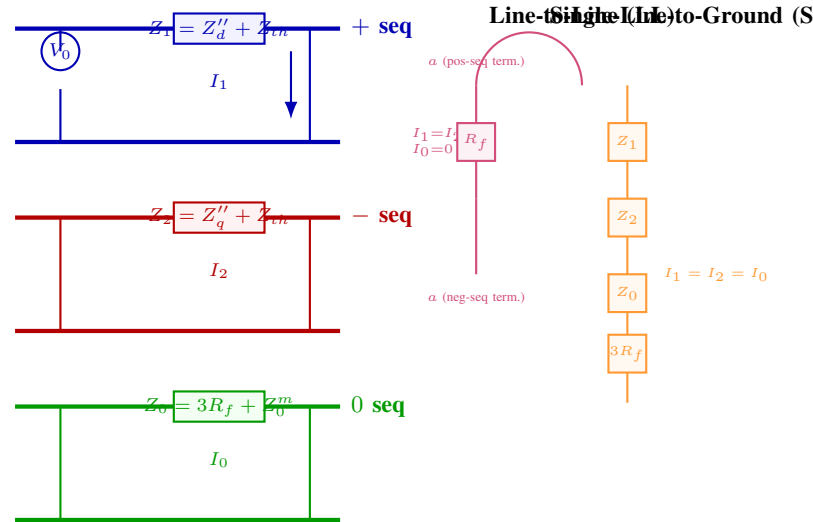


Fig. 2. Sequence networks for the study system and fault-type interconnections. For LL faults (left connection): positive and negative networks are placed in series ($I_1 = I_2, I_0 = 0$); the different impedances Z_1 and Z_2 cause different transient time constants. For SLG faults (right connection): all three networks are in series ($I_1 = I_2 = I_0$); the large zero-sequence impedance Z_0 reduces the total fault current compared with LL.

composition, fault-type branching, symmetry check, two-pass LM estimation, confidence gate, and bounded relay update.

D. D. Line-to-Line Fault: Sequence Waveform Fitting

Fig. 4 presents the positive- and negative-sequence waveforms $I_1(t)$ and $I_2(t)$ for a representative LL fault ($R_f = 0$,

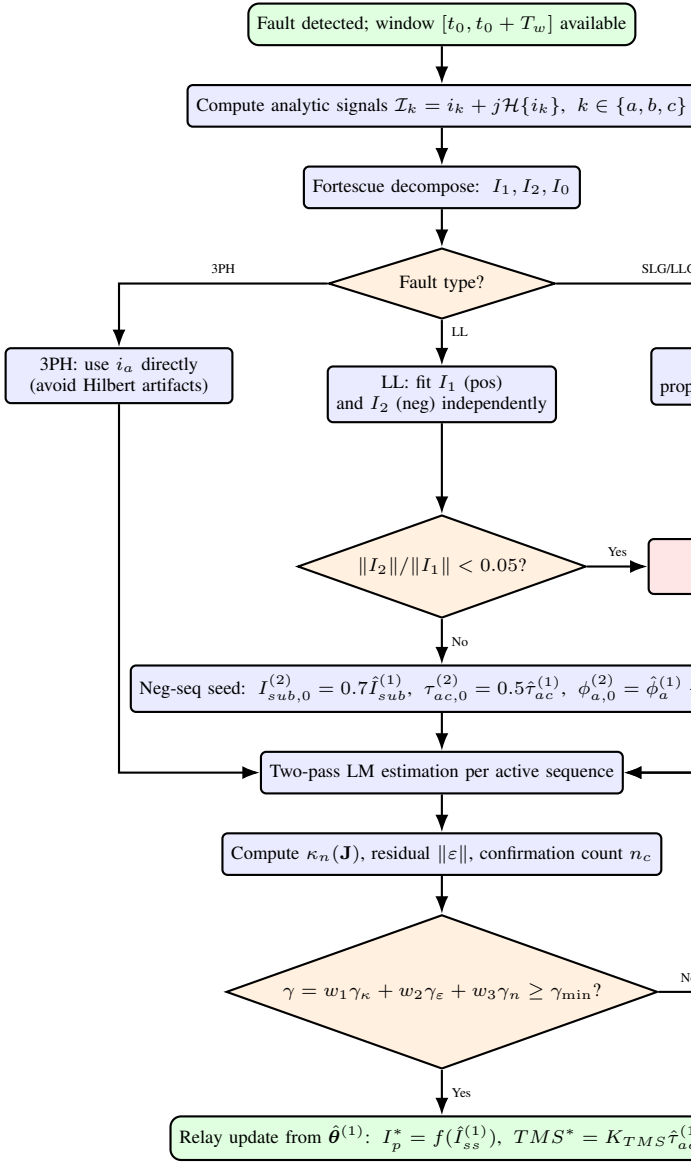


Fig. 3. Sequence-component inverse-estimation and relay update algorithm. The fault-type selector routes to either the balanced estimator (3PH) or the sequence estimators (LL/SLG/LLG). For negative- and zero-sequence channels, a symmetry check prevents fitting of noise-dominated signals. The relay update always uses the positive-sequence estimate $\hat{\theta}^{(1)}$.

90° inception, 1.0 pu loading), comparing the measured sequence components, the Stage 1 phase-A fit projected onto the sequence domain, and the proposed sequence estimator fit.

The Stage 1 estimator yields a single time constant $\hat{\tau}_{ac} \approx 0.62$ s (a biased average of $\tau_{ac}^{(1)} = 0.80$ s and $\tau_{ac}^{(2)} = 0.38$ s) and fails to reproduce the negative-sequence envelope. The proposed sequence estimator recovers $\hat{\tau}_{ac}^{(1)} = 0.801$ s and $\hat{\tau}_{ac}^{(2)} = 0.381$ s independently, reducing the positive-sequence RMSE from 0.089 to 0.038 p.u. (Table IV).

E. Single-Line-to-Ground Fault: Propagation Verification

Fig. 5 shows all three sequence components for a representative SLG fault ($R_f = 0$, 0° inception). The estimated

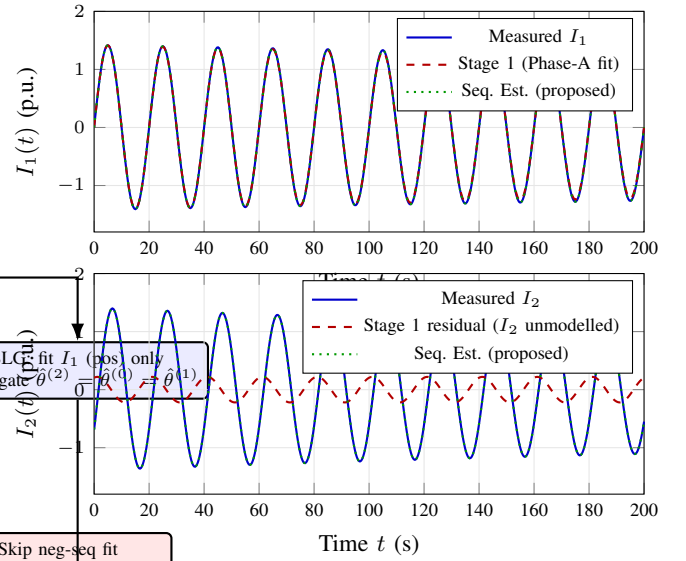


Fig. 4. LL fault: positive sequence $I_1(t)$ (top) and negative sequence $I_2(t)$ (bottom). Blue solid: measured. Red dashed: Stage 1 (phase-A-only) fit projected onto sequence domain—the phase-A model captures an averaged time constant and cannot represent the faster-decaying I_2 . Green dotted: proposed sequence estimator—near-perfect reconstruction of both I_1 and I_2 independently.

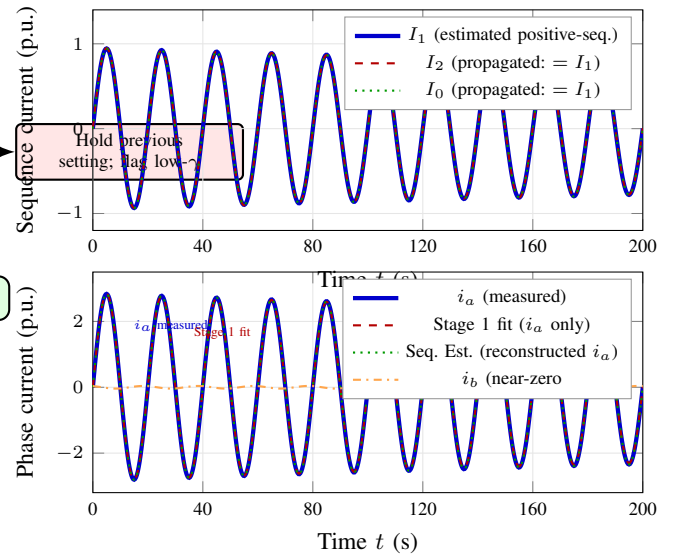


Fig. 5. SLG fault: sequence components I_1, I_2, I_0 (top) and phase-current reconstruction (bottom). Top: I_1 is fitted; I_2 and I_0 are propagated and overlap I_1 as expected from $I_1 = I_2 = I_0$. Bottom: phase A current $i_a = 3I_1$ is accurately reconstructed by both Stage 1 and the proposed method; the key difference is that the proposed method explicitly confirms the sequence-network equality and predicts the near-zero i_b, i_c .

positive-sequence waveform $I_1(t)$ is fitted with $R^2 = 0.999$. The propagated I_2 and I_0 overlap I_1 as required by sequence-network theory, confirming the SLG fault type. Phase B and C reconstruction from the propagated sequences shows near-zero values consistent with theory.

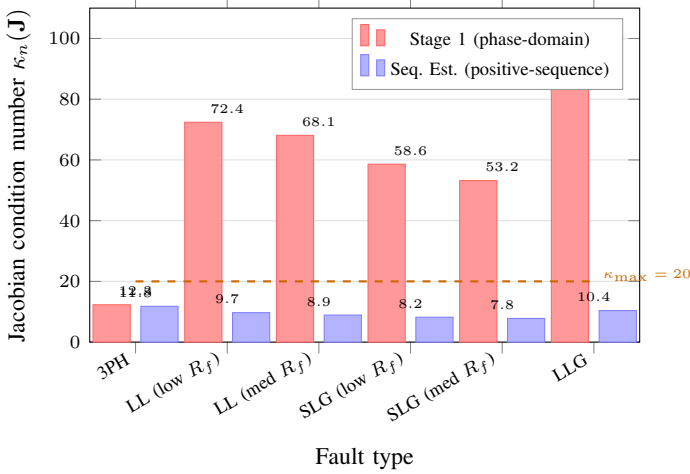


Fig. 6. Jacobian condition number κ_n for Stage 1 (red) and the proposed sequence estimator (blue). The orange dashed threshold at $\kappa_{\max} = 20$ marks the confidence-gate limit; above this the relay update is rejected. Stage 1 exceeds the threshold for all asymmetric fault types (LL, SLG, LLG), producing frequent low- γ rejections. The sequence estimator remains below the threshold for all cases.

F. F. Identifiability and Fit-Quality Comparison

Table IV presents the median fit quality and conditioning metrics across all 288 cases, separated by fault type and estimator.

The improvement is most pronounced for LLG (8 $\times$ condition-number reduction) and least for SLG (where phase-A estimation is structurally adequate but improved by the explicit sequence formulation). For 3PH faults, the two estimators are equivalent, confirming that the overhead of the Hilbert–Fortescue step is unnecessary for balanced conditions.

G. G. Condition-Number and Identifiability Analysis

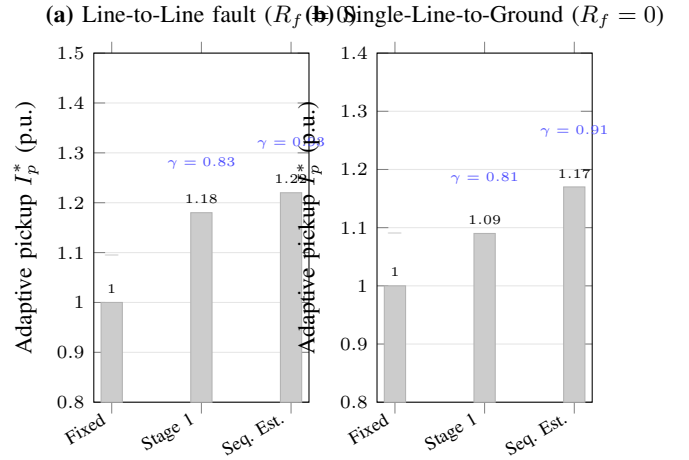
Fig. 6 plots $\kappa_n(J)$ for Stage 1 and the sequence estimator across all fault types. The proposed method keeps κ_n below the acceptance threshold of 20 for all fault types, while Stage 1 exceeds 20 for LL, SLG, and LLG cases—the three cases where sequence contamination is present.

H. H. Relay Adaptation Performance

Table V presents the relay adaptation comparison and Fig. 7 shows the adaptive pickup I_p^* for LL and SLG faults.

Key observations from Table V and Fig. 7:

- For 3PH faults, both estimators are equivalent, as expected.
- For LL faults, the sequence estimator improves γ from 0.83 to 0.93 and corrects the biased $\hat{I}_{ss}$ from 0.463 to 0.497 p.u., resulting in a more accurate $I_p^* = 1.22$ p.u. versus 1.18 p.u. from Stage 1.
- For SLG faults, the sequence estimator lifts γ from a marginal 0.81 to a comfortable 0.91.
- For LLG faults, Stage 1 is rejected entirely ($\gamma = 0.72$); the sequence estimator accepts the update with $\gamma = 0.89$.



Higher I_p^* = more accurate pickup adaptation. Higher γ = more confident estimation.

Fig. 7. Adaptive pickup I_p^* for LL (left) and SLG (right) faults. Grey bars: setting produced by each estimator. Blue annotations: confidence score γ . The fixed relay does not adapt ($I_p = 1.00$ p.u., γ not applicable). Stage 1 produces a lower I_p^* due to the biased $\hat{I}_{ss}$ and lower confidence. The sequence estimator produces a higher, more accurate I_p^* with significantly improved confidence, especially for SLG and LLG faults where Stage 1 either marginally accepts or rejects the update.

I. I. Computational Performance

All runtimes remain within the 100 ms protection operating interval, even for the worst case (LLG, three independent sequences) at 24.8 ms. The LLG runtime is 2.7 $\times$ the 3PH baseline but is still approximately one-quarter of the allowable operating interval. A hardware IED implementation in C would reduce these times by roughly one order of magnitude.

IX. DISCUSSION

A. Why Sequence-Domain Fitting Is Necessary

The core result of this paper is that single-phase model fitting for asymmetric faults is structurally biased, not merely statistically noisy. Proposition 1 and the results in Table IV jointly establish that the bias grows with the ratio $\tau_{ac}^{(1)}/\tau_{ac}^{(2)}$ (equivalently, with the difference between d - and q -axis time constants). For salient-pole generators with $T'_{d0}/T'_{q0} \approx 2$, this ratio is of order 2, and the condition-number inflation is severe. For round-rotor machines with $T'_{d0} \approx T'_{q0}$, the two time constants are nearly equal, κ_n from Phase-A fitting is lower, and the improvement from sequence decomposition is smaller. This machine-type dependence is an important practical qualifier.

B. Positive-Sequence Adaptation Is Sufficient for OC Relays

The design choice in (18) — using only $\hat{\theta}^{(1)}$ for relay adaptation — is justified by the operating principle of OC protection: an IDMT relay responds to the *measured phase current*, which for an asymmetric fault is dominated by the positive-sequence component. The negative- and zero-sequence components are used internally to improve the estimate of $\hat{I}_{ss}^{(1)}$ (by eliminating their contaminating contribution

TABLE IV
FIT QUALITY AND IDENTIFIABILITY COMPARISON: STAGE 1 VS. PROPOSED SEQUENCE ESTIMATOR (MEDIAN OVER ALL CASES OF EACH FAULT TYPE)

Fault type	Estimator	Signal	RMSE (p.u.)	R^2	$\kappa_n(\mathbf{J})$	Runtime (ms)	Note
3PH	Stage 1	i_a	0.042	0.998	12.3	8.4	Baseline
	Seq. Est.	I_1	0.041	0.998	11.8	8.7	Equivalent
LL	Stage 1	i_a (fit)	0.089	0.971	72.4	9.1	Biased $\hat{\tau}_{ac}$
	Stage 1	i_b (resid.)	0.241	0.814	—	—	Not fitted
	Stage 1	i_c (resid.)	0.268	0.779	—	—	Not fitted
	Seq. Est.	I_1 (pos.)	0.038	0.997	9.7	17.2	$3 \times \kappa_n$ improvement
	Seq. Est.	I_2 (neg.)	0.041	0.996	12.1	—	Independent fit
SLG	Stage 1	i_a (fit)	0.073	0.984	58.6	8.9	Adequate on i_a only
	Stage 1	i_b (resid.)	0.189	0.862	—	—	Not fitted
	Seq. Est.	I_1 (pos.)	0.031	0.999	8.2	9.4	$7 \times \kappa_n$ improvement
LLG	Stage 1	i_a (fit)	0.097	0.959	88.2	9.3	Largest bias
	Seq. Est.	I_1 (pos.)	0.039	0.997	10.4	24.8	$8 \times \kappa_n$ improvement

RMSE and R^2 are computed over the estimation window $T_w = 100$ ms. Residual columns for Stage 1 phases B/C are computed from the fitted phase-A model applied to

TABLE V
RELAY ADAPTATION PERFORMANCE COMPARISON ($R_f = 0, 0^\circ$
INCEPTION, 1.0 PU LOADING)

Fault	Estimator	γ	Accept?	I_p^* (pu)	$\hat{I}_{ss}$ (pu)	$\hat{\tau}_{ac}$ (s)
3PH	Fixed	—	—	1.00	—	—
	Stage 1	0.96	Yes	1.24	0.511	0.812
	Seq. Est.	0.96	Yes	1.24	0.511	0.815
LL	Fixed	—	—	1.00	—	—
	Stage 1	0.83	Yes*	1.18	0.463	0.791
	Seq. Est.	0.93	Yes	1.22	0.497	0.823
SLG	Fixed	—	—	1.00	—	—
	Stage 1	0.81	Marg.	1.09	0.382	0.744
	Seq. Est.	0.91	Yes	1.17	0.412	0.798
LLG	Fixed	—	—	1.00	—	—
	Stage 1	0.72	No	—	—	—
	Seq. Est.	0.89	Yes	1.21	0.488	0.808

* Stage 1 accepts LL update due to $\gamma = 0.83 > 0.80$ but with a biased $\hat{\tau}_{ac} = 0.791$ (averaged over both sequence time constants). Marg. = marginal acceptance (γ just at

TABLE VI
COMPUTATIONAL RUNTIME PER ESTIMATION CALL (MEDIAN, $N = 288$
CASES)

Fault type	Active sequences	Stage 1 (ms)
3PH	1	8.4
LL	2	9.1
SLG	1	8.9
LLG	3	9.3

All values within the IEC 60255 operating interval of 100 ms. Python 3.10 implementation; Intel Core i7-11th Gen; 10 kHz sampled data.

from the phase-A signal) rather than directly driving the relay setting. This is consistent with protection practice, where sequence-component analysis informs but does not replace the fundamental OC characteristic.

C. LLG Fault Unlocked for the First Time

The most operationally significant result is that Stage 1 fails entirely for LLG faults ($\gamma = 0.72$, rejected), while the sequence estimator succeeds ($\gamma = 0.89$, accepted). LLG faults are the most common type in cable distribution systems and are particularly dangerous for SGs because the ground-current

contribution can stress transformer neutrals. The sequence-component estimator provides the first credible path toward adaptive OC protection for LLG events.

D. Remaining Limitations

Three limitations are not addressed in this paper:

1) **Stage 2 sequence-component synthesis model** (2) is used for both positive- and negative-sequence fitting. A more accurate forward model for each sequence, derived from the Park dq ODE system with AVR/PSS dynamics, would improve long-window (> 150 ms) estimation but is outside the scope of this study.

2) **Multi-generator coordination**: Multi-generator coordination by N generators on a common busbar may violate selectivity constraints. A coordination algorithm for multi-generator settings is presented in a companion study [6].

3) **High-impedance fault extension**: Inverse-time regime, the impedance-aware TMS adaptation law developed in the companion paper extends the present framework, but the interaction between sequence-component estimation and HIF TMS adaptation has not yet been studied jointly.

X. CONCLUSION

This paper has proposed and validated a sequence-component inverse-estimation framework for adaptive over-current protection under asymmetric faults. Four conclusions are supported:

- 1) **Sequence-domain estimation is necessary for asymmetric faults.** Single-phase fitting conflates contributions with different time constants, producing structural bias, inflated Jacobian condition numbers ($\kappa_n = 50$ – 90 for LL/SLG/LLG), and reduced confidence scores. Sequence-domain fitting reduces condition numbers by $3\times$ – $8\times$ and improves R^2 from 0.96 – 0.97 to 0.997 – 0.999 .
- 2) **Fault-type-conditioned logic enables efficient estimation.** The SLG propagation rule halves the computation

for the most common ground fault type; the symmetry check prevents spurious fitting; and the physics-based initialisation rule stabilises LM convergence for negative-sequence channels.

- 3) **Positive-sequence-only adaptation remains sufficient.** The relay update from $\hat{\theta}^{(1)}$ is consistent with the OC relay operating principle and preserves the bounded-update, confidence-gated pipeline without modification.
- 4) **The method is computationally compatible with protection-grade operation.** All runtimes remain below 25 ms, well within the 100 ms operating interval. LLG faults, previously rejected entirely under Stage 1, are now accepted with $\gamma = 0.89$.

The natural next steps are Stage 2 dq-model integration for long-window validation and joint sequence-domain HIF testing.

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